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A Systematic Review and Benchmarking of Subthreshold / Weak-Inversion CMOS Inverter-Based Ultra-Low-Power Amplifiers: Performance Trade-Offs and Design Guidelines

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Abstract. Ultra-low-power analog front-end circuits are a key enabling technology for energy-constrained applications such as wearable electronics, wireless sensor nodes, and implantable biomedical systems. In these scenarios, aggressive scaling of supply voltage and bias current severely limits the applicability of conventional strong-inversion complementary metal–oxide–semiconductor (CMOS) analog design techniques. Operating metal–oxide–semiconductor (MOS) transistors in the subthreshold or weak-inversion region provides significantly enhanced transconductance efficiency, enabling signal amplification at extremely low power consumption. Many reported performance figures in the literature are obtained under widely varying conditions and definitions making comparison and design-oriented evaluation elusive. This paper presents a systematic review and benchmarking study of subthreshold and weak-inversion CMOS inverter-based ultra-low-power amplifiers. A structured literature screening methodology and a grouped benchmarking framework are established to extract and compare key performance metrics, including power consumption, voltage gain, bandwidth (or gain–bandwidth product, GBW), noise performance, and robustness. Based on comparative analysis under grouped operating conditions, fundamental performance trade-offs are identified, and practical design guidelines are derived for ultra-low-power analog front-end design.

Keywords: Inverter-based amplifier, Ultra-low power, Weak inversion, Gain–bandwidth product, Noise

1. Introduction

Ultra-low-power analog front-end circuits are essential components in energy-constrained systems such as wearable devices, wireless sensor nodes, and implantable biomedical electronics. These applications favour long-term autonomy over high speed, thereby constraining both power consumption and supply voltage. Conventional CMOS analog design techniques derived from strong-inversion operation face the challenges of limited voltage headroom and decreasing transconductance efficiency in aggressive voltage and current scaled regimes, restricting their applicability in the ultra-low power regime [1].

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Operation in the subthreshold or weak-inversion region provides a fundamentally different design approach, enabling amplification at nanoampere or picoampere current levels due to the high transconductance efficiency of MOS devices in this regime [2]. Among weak-inversion amplifier architectures, inverter-based designs have gained significant attention because they exploit complementary n-channel metal–oxide–semiconductor (NMOS) and p-channel metal–oxide–semiconductor (PMOS) devices to achieve higher effective transconductance while maintaining simple structure and low-voltage compatibility [3].

For example, ARE (bandwidth, GBW, etc.) are obtained under different operating conditions and often different definitions [4]. In this paper, we conduct a review and benchmarking study that attempts to distill the engineering tradeoff relevant under comparable conditions, and proposes practical design guidelines for ultra-lowpower analog front-end.

2. Systematic review methodology and benchmarking framework

This section establishes a reproducible systematic review methodology and a unified benchmarking framework to support subsequent classification and trade-off analysis. Given the strong dependence of weak-inversion amplifier performance on operating conditions, the primary objective is not to rank designs by absolute metrics but to construct a comparison space in which engineering-relevant trends can be extracted with minimal ambiguity.

2.1. Research scope and screening rules

The scope of this review is limited to CMOS inverter-based amplifiers operating in the subthreshold or weak-inversion region and explicitly targeting ultra-low-power applications. Major academic databases were queried using combinations of keywords related to weak inversion, inverter-based amplification, and ultra-low-power analog circuits.

Inclusion criteria were defined to ensure comparability and engineering relevance. Only studies reporting quantitative performance metrics were included, and each design was required to specify key operating conditions such as supply voltage and biasing approach. Results obtained through simulations or actual silicon measurements were accepted, as long as the testing conditions were adequately described. Papers that did not provide sufficient detail to interpret the metrics reported were removed from consideration, as incomplete reporting significantly increases the uncertainty in benchmarking results [5]. The screening process follows a reproducible multi-stage workflow that includes initial keyword filtering, abstract screening, and full-text evaluation.

2.2. Data extraction fields and metric normalization

To enable structured comparison, a unified set of data fields was defined for extraction from each selected study. In addition, performance metrics such as voltage gain, bandwidth, gain–bandwidth product, power consumption, and noise were recorded according to the definitions provided in the original publications.

Due to inconsistent reporting practices, direct normalization of all metrics is not always feasible. Therefore, explicit rules were established for handling incompatible definitions. When performance metrics could not be mapped to a common definition, designs were grouped according to comparable operating assumptions rather than forced into a single numerical scale [4].

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3. Classification framework: organizing literature by design dimensions

To systematically organize the diverse inverter-based weak-inversion amplifier designs reported in the literature, this review adopts a three-dimensional classification framework based on biasing strategy, feedback, gain-setting method, and circuit topology. These dimensions are selected because they precisely dictate operating-point stability, signal transfer characteristics, and prevention against ultra-low-power constraints. Herein we focus on how choices translate into a system-level trade-off, as opposed to schematic-level differences [6].

By mapping each design onto the same design dimensions, recurring architectural patterns can be identified and related to their typical benefits and costs [1]. As a result, the classification framework supports not only literature organization but also early-stage architectural decision-making.

3.1. Biasing strategies: self-biasing and external biasing

Biasing strategy determines the operating point of inverter-based weak-inversion amplifiers. It has a first-order impact on power consumption, stability, and sensitivity to process, voltage, and temperature (PVT) variations. Self-biased designs define their operating point using inherent device characteristics and avoid additional bias circuitry, thereby saving power and silicon real-estate. This methodology is appealing in applications having very little power available to them and where variable performance is tolerable [1].

However, self-biasing significantly amplifies sensitivity to device mismatch and threshold voltage variation. In weak inversion, small parameter shifts can cause large changes in drain current, resulting in unstable gain and bandwidth across PVT corners. Startup behavior is also a concern at ultra-low supply voltages, where leakage currents and parasitic effects may prevent convergence to the intended operating point [7]. These limitations limit the use of self-biased designs in applications requiring predictable, long-term, stable operation.

External biasing schemes mitigate these issues by enforcing well-defined bias currents or voltages using dedicated reference circuits. Although this increases circuit complexity and power consumption, it substantially improves reproducibility and long-term stability. Numerous reported designs demonstrate that the additional power overhead associated with bias generation is acceptable when robustness and controllability are prioritized, particularly in system-level integrations [8].

3.2. Feedback and gain setting: resistors, pseudo-resistors, and alternatives

Feedback implementation directly determines closed-loop gain, linearity, and stability. Conventional resistive feedback provides accurate, temperature-stable gain but incurs area overhead that can be significant in ultra-low-power designs. Such approaches are thus typically reserved for applications where the gain in accuracy outweighs area constraints [9].

MOS pseudo-resistors are widely adopted owing to their ability to realize extremely large resistance values with minimal area, making them highly suitable for ultra-low-frequency applications. However, their nonlinear current–voltage characteristics and sensitivity to process variation degrade gain accuracy and noise performance. These constraints are especially severe for weak-inversion operation, however, as the device characteristics are functionally strong non-linear function of bias conditions [9]. Alternative approaches of feedback, such as switched-capacitor or dynamic feedback serve to increase robustness, but these approaches will often incur additional complexity, clocking, and switching noise [10].

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3.3. Topological variants: single-stage, multi-stage, and buffered designs

Topology selection strongly influences achievable gain, bandwidth, and load-driving capability. Single-stage inverter-based amplifiers offer maximum energy efficiency and minimal area but are limited in open-loop gain and output swing. Multi-stage configurations improve gain but introduce additional poles, increasing sensitivity to mismatch and stability challenges in weak inversion [3].

Output buffering enhances load isolation and bandwidth predictability under varying load conditions, at the cost of increased power consumption and noise contribution. Benchmarking results show that buffered designs are particularly beneficial when interfacing with capacitive or poorly defined loads, despite their higher power consumption [11]. Consequently, topology choice reflects a balance between energy efficiency and system integration requirements, rather than a universally optimal solution.

4. Benchmarking results and performance trade-offs

Using the unified data fields and comparability grading defined in Section 2, benchmarking was conducted across different design categories to extract reusable engineering trade-offs. Rather than ranking designs by absolute performance metrics, the analysis focuses on identifying dominant trends and constraint-driven behaviors that persist across comparable operating conditions. This approach emphasizes engineering insight over record-oriented comparison.

4.1. Metric distribution and comparability issues

Reported power consumption spans several orders of magnitude, ranging from nanowatt-level designs optimized for extreme energy efficiency to microwatt implementations targeting improved bandwidth or robustness. Voltage gain and bandwidth exhibit similarly wide distributions across the surveyed literature. A significant portion of this variation originates from differences in test conditions, such as supply voltage, load capacitance, and gain configuration, rather than intrinsic circuit capability [4].

By grouping designs according to comparability level, it becomes possible to separate architectural trends from reporting artifacts. Designs within the same group exhibit clearer performance envelopes that reflect underlying physical constraints, whereas cross-group comparisons often exaggerate differences driven primarily by operating assumptions. This observation highlights the limitations of direct numerical ranking and reinforces the importance of context-aware benchmarking in weak-inversion amplifier evaluation.

4.2. Power–bandwidth (or GBW) trade-offs

Within comparable groups, a consistent power–bandwidth trade-off is observed. In weak inversion, transconductance scales approximately with bias current, while parasitic capacitances remain relatively constant, imposing a fundamental limit on achievable bandwidth [1]. As a result, increasing bandwidth inevitably increases power consumption, regardless of the specific circuit implementation.

For example, externally biased or buffered designs may achieve higher bandwidth at a given power level than self-biased single-stage architectures. Still, they do not alter the underlying trade-off slope.

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4.3. Power–gain trade-offs and gain-setting impact

Closed-loop gain selection significantly influences the relationship between power consumption and achievable gain. Multi-stage designs and aggressive feedback techniques increase gain but often reduce bandwidth and robustness. Benchmarking results indicate strong coupling among gain, linearity, and stability in weak-inversion operation, particularly as designs approach the minimum-bias-current limit [8].

Furthermore, gain-setting methods affect sensitivity to process variation and noise. Designs that prioritize high gain through increased stage count or strong feedback frequently exhibit degraded tolerance to mismatch and PVT variation. These trends emphasize that gain enhancement in weak inversion must be evaluated alongside its impact on stability and reproducibility, rather than as an isolated objective.

4.4. Noise and robustness: key bottlenecks

Noise performance and robustness to PVT variation emerge as primary bottlenecks for practical deployment. Many publications either omit noise characterization or report noise metrics under incompatible conditions, limiting meaningful comparison across designs [12]. This lack of standardized noise reporting obscures an important performance dimension, particularly in low-frequency sensing applications.

Benchmarking trends indicate that design choices such as moderate increases in bias current, external biasing, and controlled feedback significantly improve robustness and noise behavior, albeit at a power cost [13]. These results suggest that absolute minimum power consumption is rarely the optimal operating point for real-world systems and that modest power trade-offs can yield disproportionate gains in usability and reliability.

5. Design guidelines

Based on the classification and benchmarking results, practical and actionable design guidelines can be formulated for inverter-based weak-inversion amplifiers. These guidelines are derived from recurring patterns observed across the reviewed literature. They are intended to support early-stage architectural decisions by linking application priorities to appropriate combinations of biasing strategy, feedback implementation, and topology [8].

For applications prioritizing minimum power consumption, self-biased single-stage inverter-based designs represent the most energy-efficient choice. Such architectures minimize bias circuitry and silicon area, enabling operation at extremely low current levels. However, the literature consistently shows that this approach results in increased sensitivity to process, voltage, and temperature variations, as well as reduced predictability of gain and bandwidth [14]. As a result, self-biased designs are most suitable for applications with relaxed accuracy requirements and stable operating environments.

When low noise or high robustness is a primary requirement, externally biased architectures with controlled feedback are generally preferable. Explicit biasing improves operating-point stability and reduces sensitivity to device mismatch, while well-defined feedback paths enhance gain controllability and linearity. Several reported designs demonstrate that the additional power overhead introduced by bias circuitry is modest compared to the resulting improvement in noise performance and reproducibility [12].

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Topology and feedback choices should be made jointly, with explicit consideration of system-level constraints. Single-stage inverter-based amplifiers offer maximum energy efficiency but limited gain and load-driving capability. Multi-stage or buffered architectures can address these limitations, but at the cost of increased power consumption, noise accumulation, and stability challenges [11]. Benchmarking results indicate that buffered or multi-stage solutions are most effective when higher gain or improved load isolation is required.

Finally, designers are encouraged to avoid optimizing individual metrics in isolation. Weak-inversion operation trades power consumption, gain, bandwidth, noise, and robustness against one another. For many systems, modest increases in bias current together with a careful increase in architectural complexity will lead to disproportionally large gains in stability, long-term usability, and so on. The absolute minimum power consumption thus becomes an unattainable target [13].

6. Conclusion

This paper has presented a comprehensive systematic review and benchmarking study of subthreshold and weak-inversion CMOS inverter-based ultra-low-power amplifiers. A reproducible literature screening methodology and grouped comparison framework were established to address inconsistencies in reported performance metrics. By organizing representative designs according to biasing strategy, feedback implementation, and topology, fundamental trade-offs among power consumption, gain, bandwidth, noise, and robustness were identified.

The analysis demonstrates that inverter-based weak-inversion amplifiers achieve exceptional energy efficiency at extremely low bias currents. Still, this efficiency is accompanied by limited controllability of bandwidth and increased sensitivity to process, voltage, and temperature variations. Noise performance and robustness thus emerge as primary bottlenecks for practical deployment, particularly in precision sensing and long-term autonomous systems where stable operation, repeatability, and predictable lifetime behavior are critical design requirements.

Importantly, the benchmarking results show that many apparent performance advantages arise from differences in operating conditions rather than intrinsic circuit superiority. When comparability boundaries are enforced, performance trends converge along clear physical limits governed by weak-inversion device behavior. This highlights the need to interpret reported metrics within their specific evaluation context rather than treat them as universally applicable performance indicators, especially when transferring circuit concepts across applications or technology nodes. This work delineates circuit-system trade-offs for ultra-low-power analog front-ends, offering designers a structured reference. It demonstrates that balanced operating points, not minimal-power extremes, optimize overall system performance. These guidelines support robust design decisions and prioritize future research on reproducible benchmarking and application-aligned optimization.

References

[1] Vittoz, E. A. (2009). Weak inversion for ultra low-power and very low-voltage circuits. In 2009 IEEE Asian Solid-State Circuits Conference. IEEE. pp. 129-132.
[2] Soeleman, H., Roy, K. (1999). Ultra-low power digital subthreshold logic circuits. In Proceedings of the 1999 international symposium on Low power electronics and design. pp. 94-96.
[3] Manfredini, G., Catania, A., Benvenuti, L., Cicalini, M., Piotto, M., Bruschi, P. (2020). Ultra-low-voltage inverter-based amplifier with novel common-mode stabilization loop. Electronics, 9(6): 1019.
[4] Grasso, A. D., Pennisi, S., Venezia, C. (2023) A survey of ultra-low-power amplifiers for internet of things nodes. Electronics, 12(20): 4361.
[5] Kitchenham, B. (2004). Procedures for performing systematic reviews. Keele University.

DOI: 10.54254/2755-2721/2026.BJ33823

[6] Lv, L., Zhou, X., Qiao, Z., Li, Q. (2019) Inverter-based subthreshold amplifier techniques and their application in 0.3-V $\Delta\Sigma $-modulators. IEEE J. Solid-State Circuits, 54(5): 1436-1445.
[7] Wang, A., Chandrakasan, A. (2005) A 180-mV subthreshold FFT processor using a minimum energy design methodology. IEEE J. Solid-State Circuits, 40(1): 310-319.
[8] Yeknami, A. F., Qazi, F., Alvandpour, A. (2013) Low-Power DT $\Delta\Sigma $ Modulators Using SC Passive Filters in 65 nm CMOS. IEEE Trans. Circuits Syst. I Regul. Pap., 61(2): 358-370.
[9] Guglielmi, E., Toso, F., Zanetto, F., Sciortino, G., Mesri, A., Sampietro, M., Ferrari, G. (2020) High-value tunable pseudo-resistors design. IEEE J. Solid-State Circuits, 55(8): 2094-2105.
[10] Martincorena-Arraiza, M., Carlos, A., Carlosena, A., Lopez-Martin, A. (2022). AC coupled amplifier with a resistance multiplier technique for ultra-low frequency operation. Int. J. Electron. Commun, 149: 154176.
[11] Della Sala, R., Centurelli, F., Scotti, G. (2022) A novel differential to single-ended converter for ultra-low-voltage inverter-based OTAs. IEEE Access, 10: 98179-98190.
[12] Harrison, R. R., Charles, C. (2003) A low-power low-noise CMOS amplifier for neural recording applications. IEEE J. Solid-State Circuits, 38(6): 958-965.
[13] Akbari, M., Hashemipour, O. (2017) A 63-dB gain OTA operating in subthreshold with 20-nW power consumption. Int. J. Circ. Theory Appl., 45(6): 843-850.
[14] Roy, A. (2017). Power Management in Ultra-Low Power Systems (Doctoral dissertation), University of Virginia.